Helical Coding Challenge

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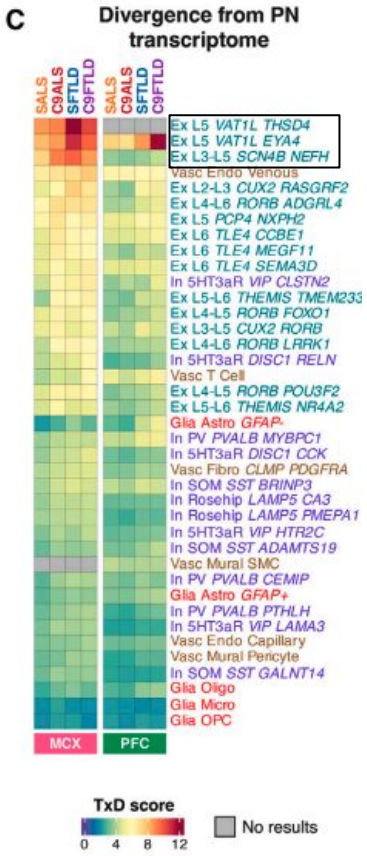
Cell Types and Gene Selection



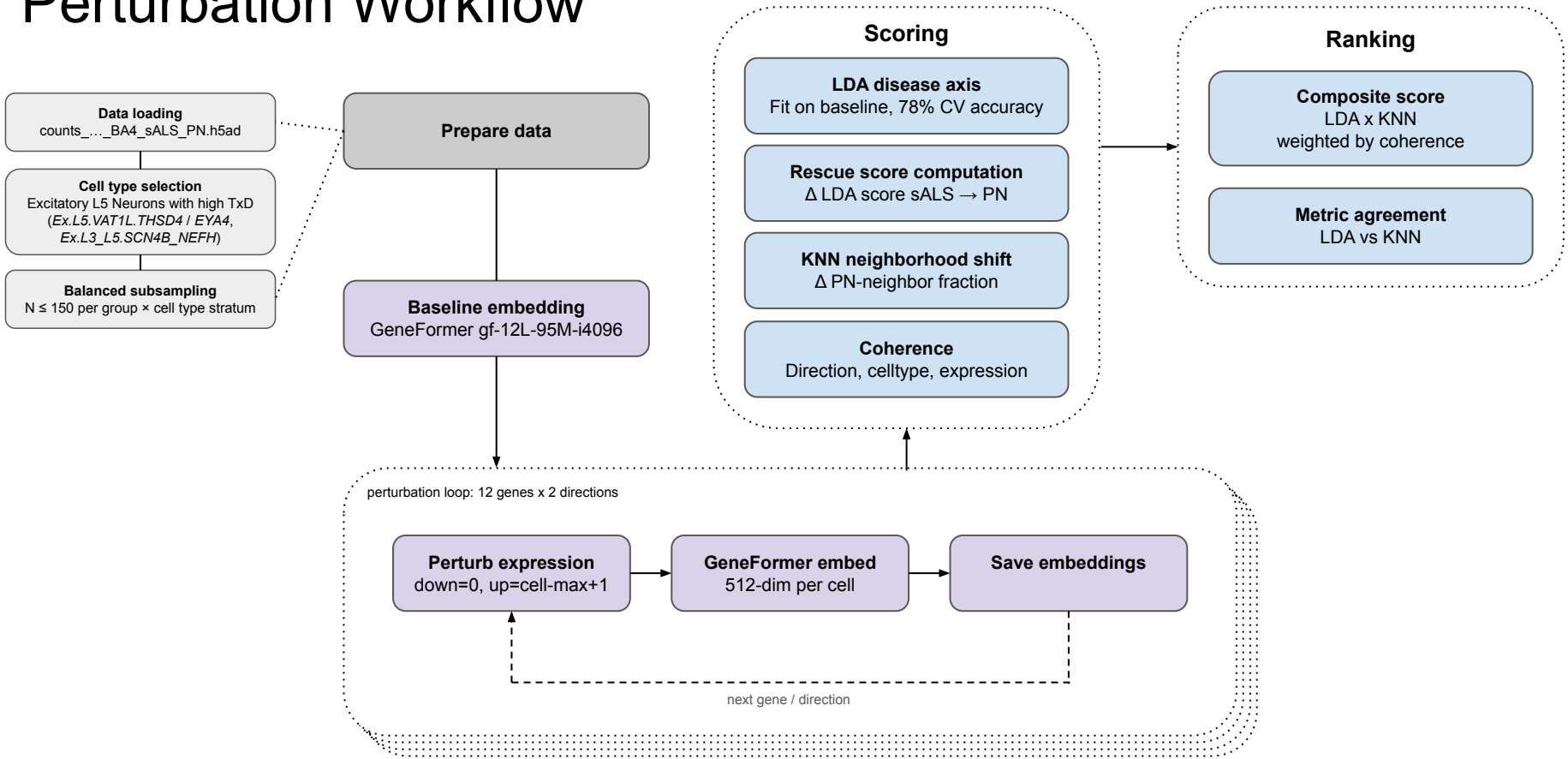
<u>Celltypes</u>

- Excitatory L5 populations
- High transcriptional divergence between sALS and healthy cells

<u>Genes</u>

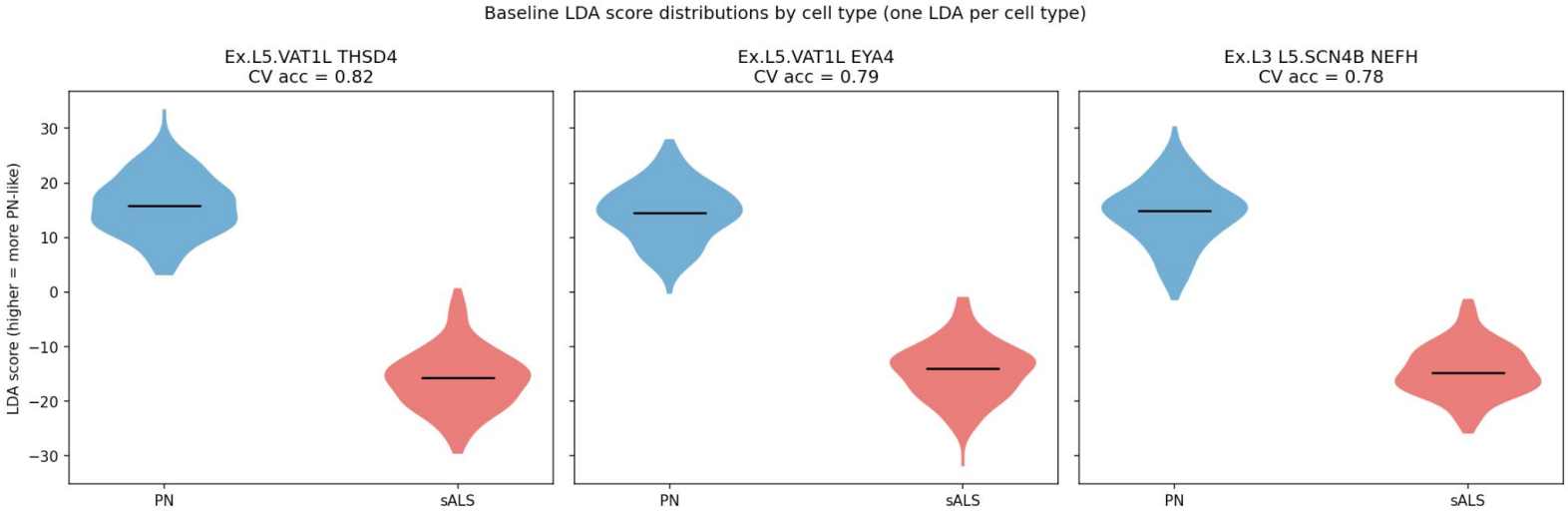
- Predicted master regulators of disease state: *MYT1L, REST, SREBF2*
- Top Drivers of Transcriptional Divergence: *LYNX1, TBR1*
- Axonal / Structural Genes Correlated with Divergence: *KIF5A, DCTN1, DYNC1H1, TUBA4A*
- Highly Specific Differentially Expressed Genes: *NEFH, TTC14, HSP90AA1*

Perturbation Workflow



Results

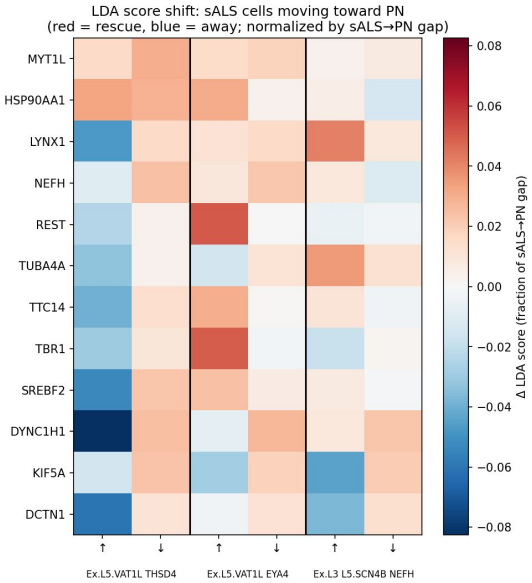
The disease state is linearly encoded in the Geneformer embedding

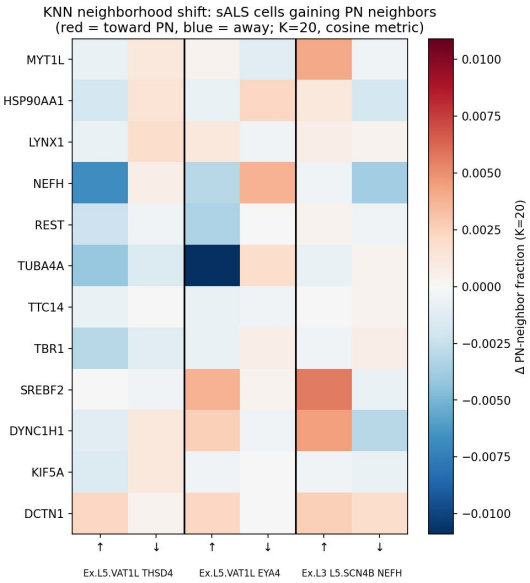


- GeneFormer, without any disease-specific fine-tuning, encodes disease state as a linearly separable direction in its 512-dimensional embedding space.
- LDA finds the disease axis because it's told to. UMAP and clustering aren't, so they drown it in noise.

Results

LDA and KNN metrics don’t agree



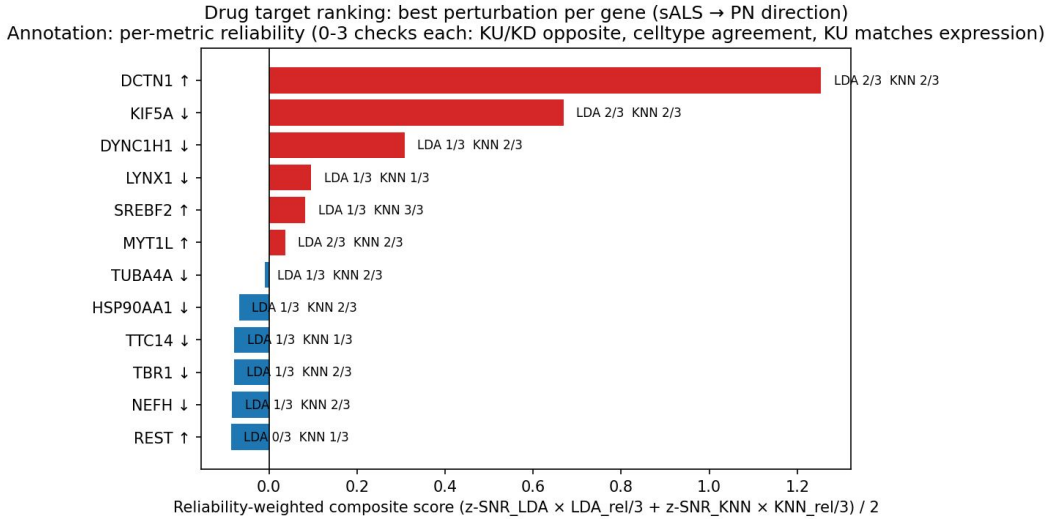


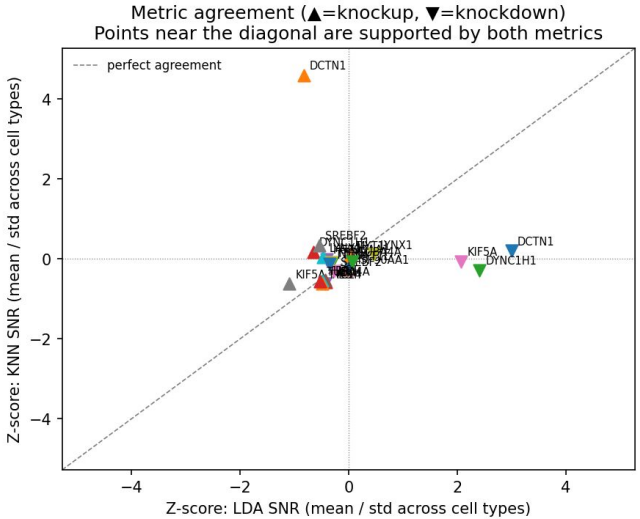
- Small effect sizes
- Low cross-celltype consistency
- KU / KD often agree
- Directions often contradict expression expectations
- The two metrics largely disagree with each other as well, suggesting they are operating at the noise floor

→ Coherence score

Results

Ranking





- Composite score = (z(SNR_LDA) × coh_LDA + z(SNR_KNN) × coh_KNN) / 2
  - SNR_m = mean(shift) / std(shift)
  - Z-score both SNRs across all (gene, direction) pairs
  - Weight each z-score by its metric's coherence score

- Low metric agreement

Conclusions

Main finding

- **The disease axis exists** and is clearly detectable via LDA (~78% CV accuracy), but single-gene rank perturbations produce no detectable movement along it.

Why perturbations fail

- **GeneFormer encodes global transcriptional state.** A single-gene rank change is a tiny perturbation to a 22,000-gene ordering. The model is robust to this as it does not reflect actual biology.

- **ALS divergence is a network-level phenomenon.** The sALS/PN gap reflects hundreds of genes and regulatory interactions. Perturbing one gene in vivo typically triggers a downstream cascade, the rank-shift approach misses this entirely.

- **ALS has a post-translational dimension.** TDP-43 mislocalization and cytoplasmic aggregation, the hallmark pathology in ~97% of ALS cases, is post-translational and entirely invisible to transcriptomic approaches. A model trained on mRNA cannot detect or predict this layer of disease mechanism, regardless of perturbation strategy.

What would help

- **Perturbation- and GRN-aware models** (*GEARS*, *scGPT*, *CellOracle*) predict transcriptional cascades and produce biologically realistic post-perturbation states, addressing the network problem.

- **Disease-state fine-tuning of GeneFormer** would make the disease axis a primary direction of variation in the latent space, potentially amplifying weak perturbation signals, addressing the model robustness problem.

- **GRN-guided multi-gene perturbation.** Use a directed TF-target GRN to propagate each perturbation to downstream targets, weighted by regulatory distance, then shift all nodes simultaneously.